$\boxed{1}$

Analog signal $\xrightarrow{\text{sampling}}$ Discrete signal.

$\downarrow$ Quantization. Binarization

Digital signal.

$\boxed{2}$ $101 + (-74)$.

1's complement

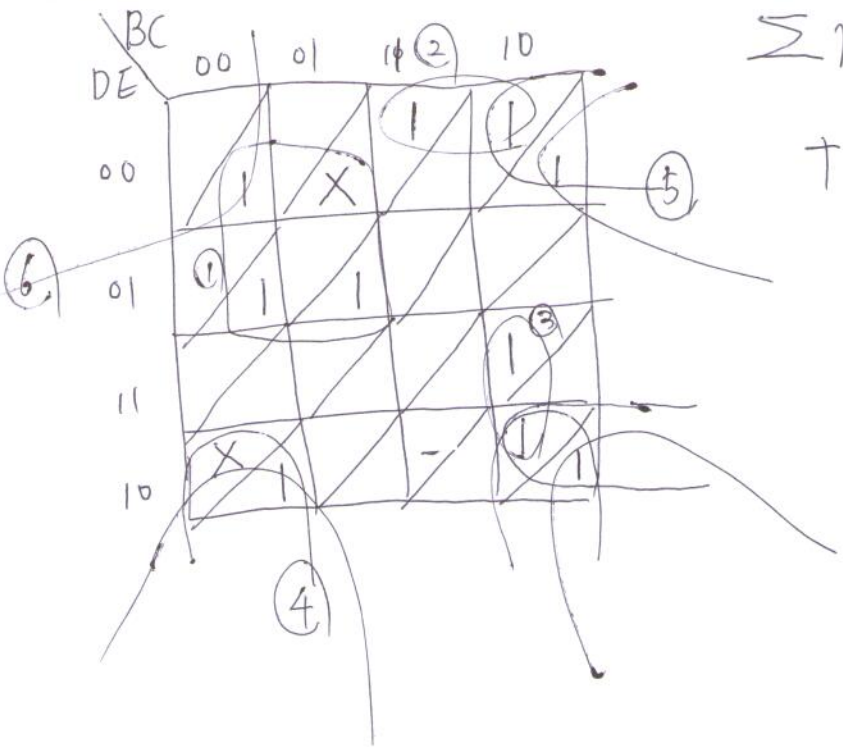
$$
\begin{array}{r}
01100101 \\
+)\ 10110101 \\
\hline
1\ \ 00011010 \\
\hline
00011011
\end{array}
$$

2's complement

$$
\begin{array}{r}
01100101 \\
+)\ 10110110 \\
\hline
\boxed{1}\ 00011011
\end{array}
$$

discard.

3 A 1/0


BC
DE
00
01
11
10
00
01
11
10
1
X
1
1
1
1
1
1
1
X
1
1
1
1
1
1
1

$\boxed{4}$.

4-bit 2's complement. $\begin{cases} A = A_3A_2A_1A_0 \\ B = B_3B_2B_1B_0 \end{cases}$

if $A > B \Rightarrow F = 1$.

1) $A_3' B_3$.

2) $(A_3 \equiv B_3)\ A_2' B_2$.

3) $(A_3 \equiv B_3)(A_2 \equiv B_2).\ A_1' B_1$

4) $(A_3 \equiv B_3)(A_2 \equiv B_2)(A_1 \equiv B_1)\ A_0' B_0$.

2-input N-XOR gate 6개, 1) 2-input AND gate : 1개

2) 3-input AND gate : 1개.

3) 4-input AND gate 1개

4) 5-input " 1개

(1) + (2) + (3) + (4) → 4-input OR gate : 1개

$\boxed{5}$ Ex-3 code $\longrightarrow$ # of 1's XY (2 bits).

Y

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | 1 | 0 | 1 |
| 01 | X | 0 | X | 0 |
| 11 | 0 | 1 | X | 1 |
| 10 | X | 0 | X | 0 |

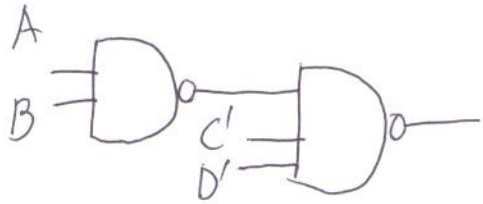
$A'C'D'$ $B'C'D'$

$BCD$ $ACD$.

X

| CD \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | 0 | 1 | 0 |
| 01 | X | 1 | X | 1 |
| 11 | 1 | 1 | X | 1 |
| 10 | X | 1 | X | 1 |

$AB$, $C$. $D$.



$$\begin{cases} m(7,\ 12,\ 14,\ 15) \\ d(1, 3, 5, 8, 10, 11, 13) \end{cases}$$

ABCD

$\boxed{6}$

| Column 1. | | Column 2. | | Column 3 | | PI. |
|---|---|---|---|---|---|---|
| 1, | 0001 ✓ | (1,3) | 00-1 ✓ | (1,3,5,7) | 0--1 | A'D |
| 8, | 1000 ✓ | (1,5) | 0-01 ✓ | (8,10,12,14) | 1--0 | AD' |
| 3, | 0011 ✓ | (8,10) | 10-0 ✓ | ~~(8,12,10,14)~~ | ~~1--0~~ | |
| 5, | 0101 ✓ | (8,12) | 1-00 ✓ | ~~(3,7,11,15)~~ | ~~--11~~ | |
| 10, | 1010 ✓ | (3,7) | 0-11 ✓ | (3,11,7,15) | --11 | CD |
| 12, | 1100 ✓ | (3,11) | -011 ✓ | (5,7,13,15) | -1-1 | BD |
| 7, | 0111 ✓ | (5,7) | 01-1 ✓ | ~~(5,13,7,15)~~ | ~~-1-1~~ | |
| 11, | 1011 ✓ | (5,13) | -101 ✓ | (10,11,14,15) | 1-1- | AC |
| 13, | 1101 ✓ | (10,11) | 101- ✓ | ~~(10,14,11,15)~~ | ~~1-1-~~ | |
| 14, | 1110 ✓ | (10,14) | 1-10 ✓ | (12,13,14,15) | 11-- | AB |
| 15, | 1111 ✓ | (12,13) | 110- ✓ | ~~(12,14,13,15)~~ | ~~11--~~ | |
| | | (12,14) | 11-0 ✓ | | | |
| | | (7,15) | -111 ✓ | | | |
| | | (11,15) | 1-11 ✓ | | | |
| | | (13,15) | 11-1 ✓ | | | |
| | | (14,15) | 111- ✓ | | | |

P. I. 6 ea.

$\boxed{6}$

PI chart.

| | 7 | 12 | 14 | 15 |
|---|---|---|---|---|
| 1. A'D | X | | | |
| 2. AD' | | X | X | |
| 3. CD | X | | | X |
| 4. BD | X | | | X |
| 5. AC | | | X | X |
| 6. AB | | X | X | X |

$$(P_1 + \underbrace{P_3 + P_4}_{\times})(\underline{P_2 + P_6})(\underline{P_2} + P_5 + \underline{P_6})(\underbrace{P_3 + P_4}_{\times} + P_5 + P_6)$$

$$= (P_2 + P_6)(P_3 + P_4 + P_1P_5 + P_1P_6)$$

$$= P_2P_3 + P_2P_4 + P_1P_2P_5 + \cancel{P_1P_2P_6} + P_3P_6 + P_4P_6 + \cancel{P_1P_5P_6} + P_1P_6$$

M.S.OP : 5가지. $\left( \begin{array}{lll} AB + A'D, & AB + CD, & AB + BD \\ AD' + CD, & AD' + BD. & \end{array} \right.$